

EEG Signal Classification of Hand Grasping States as Part of the Development of an Open Device Brain-Computer Interface

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Abstract—This study presents the classification of hand grasping states based on mu band EEG signals as part of an open-source brain-computer interface (BCI) development. EEG data were recorded using low-cost hardware consisting of an Arduino UNO and BioAmp EXG Pill, with electrodes placed on C4 and Cz in a bipolar configuration. The recorded signals were preprocessed using a bandpass filter, followed by decomposition with discrete wavelet transform (DWT) to isolate the alpha (8–13 Hz) and beta (13–30 Hz) frequency band. Power spectral density (PSD) was extracted using the Welch method and used as features for classification. A multilayer perceptron neural network was trained to distinguish between rest and grasping conditions. The model achieved satisfactory performance with consistent classification accuracy and minimal overfitting, demonstrating the feasibility of EEG-based BCI systems using affordable components. The findings support the potential of open-source platforms for accessible and personalized neurotechnology, particularly in assistive and rehabilitative applications.

Index Terms—brain-computer interface, mu rhythm, neural network, open device, motor imagery

I. INTRODUCTION

BCI or Brain-Computer Interface has been an active area of research for rehabilitation systems. It allows a direct communication between brain and hardware devices without any physical movement [1]. Since BCI needs to directly communicate with the brain, EEG (electroencephalogram) plays an important role in BCI systems. User's intention detection is done by identifying the features from EEG signal. To this day, it remains one the main challenges in developing a BCI system.

One of indicator to analyse movement intention of user is event-related desynchronization (ERD) and event-related synchronization (ERS). These are a phenomenon where the power of EEG signal in a specific range of frequency changes due to motoric activity or imagery [2]. These changes are mostly inspected in the range of 8-13 Hz or as known as mu band [3], [4]. The phenomenon is observed by local field potential (LPF) in cortical [5] and sub-cortical area [6], [7]. Since it's directly caused by the user imagining a certain

movement, this feature can represent the user's intention more intuitively.

In 2023, stroke affected approximately 8.3 out of every 1,000 individuals in Indonesia and accounted for 11.2% of all reported disability cases [8]. Primary disability that caused by stroke is motoric disorders such as muscle weakness, impaired balance and body coordination [9]. Among the affected motor functions, hand mobility is particularly crucial, as it plays a central role in performing daily activities such as eating, writing, bathing, dressing, and using the toilet [10]. Most post stroke patients have low score in Manual Muscle Testing (MMT) scale which measure the motoric ability [11]. Consequently, instead using the EMG-based system, the technology for restoration and support of hand function with BCI system have become a promising alternative in post-stroke rehabilitation.

As powerful as it sounds, BCI system with event-related EEG features has quite many setbacks. The major one is the individuality of person which cause great differences between the recorded brain activities among various people [12]. This limitation remains one of the primary barriers preventing BCI-based devices from becoming widely available at an affordable price. However, ongoing advancements in EEG hardware and the growth of open-source research initiatives offer promising opportunities to address this issue by enabling more cost-effective solutions.

This research aims to classify two conditions of hand grasping based on mu band EEG signal. By utilizing open-source hardware such as the Arduino UNO and BioAmp EXG Pill, the system was developed to be low-cost while still delivering adequate performance for EEG signal acquisition and classification tasks. To handle the complexity of the data, a neural network was designed and implemented to perform the classification. This study is also a part of a development of an open-source BCI system. The primary contribution of this study lies in its approach to signal feature extraction and classification. The experiment setup, data acquisition process, and signal processing methods are described in detail. Fur-

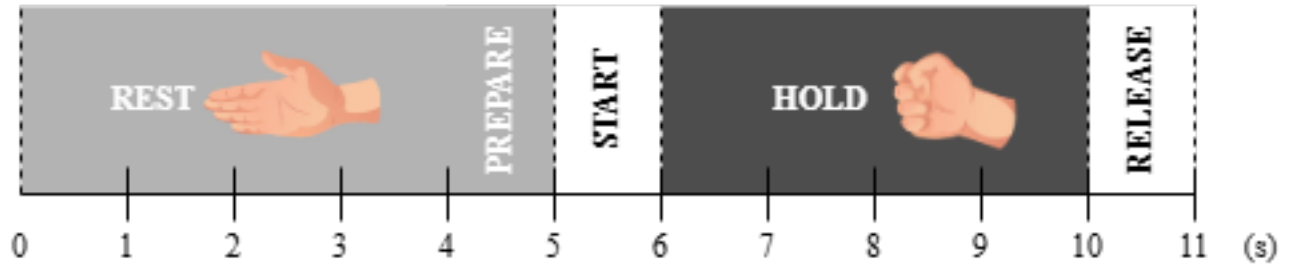


Fig. 1. Periods of hand grasp experiment.

thermore, the results of feature extraction and neural network training are thoroughly discussed.

II. METHODS

A. Participant and Experimental Environment

The experiment was carried out on a male participant aged 21-22 years. The participant was right-handed and had no history of neurological disorders. The purpose and procedure of the experiments are informed to the participant beforehand. The experiment was conducted in a room with minimal distractions. The sessions were repeated on different days to minimize participant fatigue. During the experiment, the participant was seated on a chair with a backrest, facing a screen that displayed the experimental instructions.

B. Experimental Design

To maintain participant alertness throughout the session, the experimental instructions were delivered using two types of videos. The first type consisted of main instruction videos that explained what to do and when to do it. The second type comprised interlude videos, which included short, entertaining clips selected based on the participant's preferences.

In each main instruction, the sequence consisted of four distinct periods: rest, start, hold, and release. As shown in the Fig. 1, the rest period lasted for 5 seconds. During this time, the screen displayed a black background with the word "REST" and a ticking sound played once per second. On the fourth second of the sequence, the text changed to "PREPARE." An audio cue was then played, and the text changed again to "START" as an instruction for the participant to begin grasping their left hand. The subsequent 4 seconds constituted the hold period, during which the participant was required to maintain their grasp. At the end of the sequence, another audio cue and text prompt were presented to instruct the participant to release their grasp.

C. EEG Recording

The EEG signal was recorded with bipolar configuration electrodes as shown in Fig. 2. Two electrodes were placed at positions C4 and Cz, and one additional electrode was used as the hardware ground reference, placed at A1. A gel-type electrode was used in this experiment.

For the hardware setup, an Arduino UNO was paired with the BioAmp EXG Pill as the amplifier. The BioAmp EXG

Pill module is an open device built using the *TL084CPW* operational amplifier manufactured by Texas Instruments. It operates at a 5 V supply voltage and features a high input impedance of 1- T Ω . Notably, the module is compact in size, measuring approximately 2.54 cm $\times$ 1 cm. The recommended sampling rate for this module is 512 Hz. Accordingly, the EEG signal was recorded at this rate.

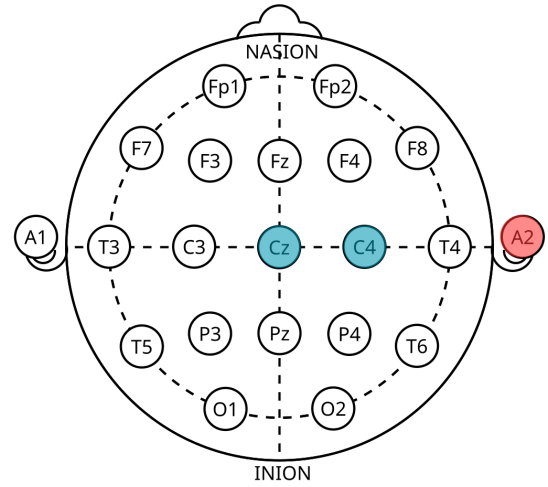


Fig. 2. EEG electrode layout.

D. Signal Processing

The recorded data are segmented into two groups according to periods. The first group is when the participant is in the REST period, and the second group is the data during the PREPARE and START periods, hereafter referred to as TASK. Because EEG signals are prone to non-stationary noise that can vary across these periods, an effective denoising method is required before performing frequency-domain analysis.

As a non-stationary signal, EEG has properties that change over time [13]. The Discrete Wavelet Transform (DWT) is a method that integrates information from both the time and frequency domains within a single analytical framework [14]. This makes DWT more advantageous than conventional band-pass filtering, which lacks sensitivity to time-localized changes within frequency bands. Although time-frequency information can also be extracted using the Short-Time Fourier Transform (STFT), the DWT provides a more flexible and efficient alternative as it overcomes the fixed resolution limitation inherent

in STFT [14]. Therefore, the Discrete Wavelet Transform is used in this study.

Signal processing was done with MATLAB. Initially, the signal was bandpass filtered using an FIR filter with a 5 and 45 Hz cutoff of frequency. This was done to eliminate or avoid components of power line interference and motion artifacts. The filtered signal was then decomposed using the discrete wavelet transform (DWT). In this study, the DWT was implemented with periodized signal extension mode ('per') and the Symlet 13 ('sym13') wavelet as the mother wavelet. The DWT decomposes the signal into time-frequency components using scaled and shifted versions of the mother wavelet defined as:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right) \quad (1)$$

where a is the scale a and b is the translation.

With a sampling frequency of 512 Hz, the decomposition levels of DWT correspond to specific frequency bands. The upper limit frequency range of the approximation coefficients at each level can be approximated by:

$$f_{\text{high}} = \frac{f_s}{2^j} \quad (2)$$

where j denotes the decomposition level and f_s is the sampling frequency. Based on this relationship, level 4 ($j = 4$) corresponds to approximately 0 to 32 Hz, and level 5 ($j = 5$) corresponds to approximately 0 to 16 Hz, aligning with the beta and mu band. This level was selected for feature extraction due to its relevance in motor-related EEG activity. The wavelet coefficients were then reconstructed as time-domain signals for further analysis.

Subsequently, power spectral density (PSD) estimation was performed using the Welch method, which averages the periodograms of overlapping segments as:

$$P_{xx}(f) = \frac{1}{L} \sum_{l=1}^L |\text{FFT}_l(f)|^2 \quad (3)$$

where L is the number of segments. The FFT was computed using 2000 points, with a Hamming window of 256 samples and an overlap of 8 samples. The sampling rate was 500 Hz. Power values within the 5 to 30 Hz range were extracted as features.

III. RESULT AND DISCUSSION

A. Feature Extraction

After undergoing filtering, decomposition, and reconstruction, the signal demonstrates a noticeable reduction in noise compared to the raw data, suggesting that the DWT method effectively isolates the relevant frequency bands. However, the event-related patterns are not yet clearly distinguishable in the time domain, as seen in Fig. 3. Therefore, power spectral analysis using the Welch method was employed to extract more prominent and interpretable features in the frequency domain.

Fig. 4 illustrates the average power spectral density of EEG signals in all data sets within the 0 to 30 Hz frequency

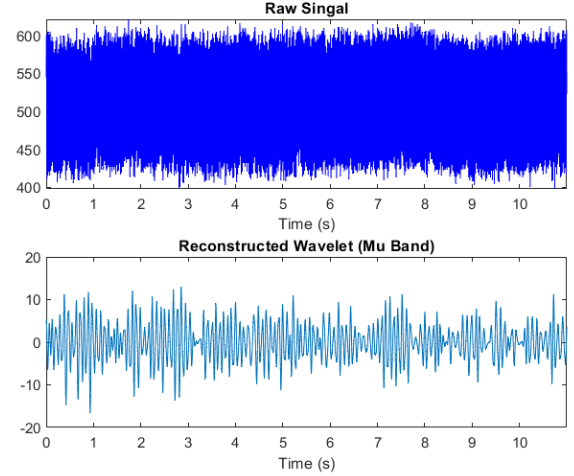


Fig. 3. Raw and reconstructed wavelet signal.

range, comparing the rest and hold conditions. Both conditions exhibit a similar peak in the 10 to 12 Hz range, which corresponds to typical mu rhythm activity. The REST period displays a higher power compared to the TASK condition, indicating the presence of event-related desynchronization (ERD). This phenomenon is observed in both the alpha and beta bands. However, the decrease in PSD is not statistically significant. This may be due to insufficient engagement in motor imagery, as the participant was instructed to actively move their hand rather than solely imagine the movement.

B. Neural Network Design and Training

Among other machine learning models, the multilayer perceptron neural network is chosen based on its ability to learn with complex datasets. A trial-and-error approach was employed to iteratively tune various parameters and identify the optimal neural network configuration. The final architecture of the neural network consists of three hidden layers with consecutively 80, 78, and 65 numbers of neurons. Each layer utilizes Rectified Linear Unit (ReLU), a widely used activation function. ReLU offers greater efficiency compared to other activation functions, as it activates only a subset of neurons at any given time rather than all neurons simultaneously [15]. To further prevent overfitting, a dropout layer with a rate of 0.15 is applied after the first hidden layer.

For training purposes, the main dataset was randomly shuffled to ensure reproducibility. Then it was partitioned into training, testing, and validation sets with proportions of 70%, 15%, and 15%, respectively. The neural network was trained for a maximum of 15 epochs to allow sufficient learning while minimizing the risk of overfitting. The use of a validation set enables performance monitoring during training, while the testing set is reserved for final model evaluation.

The training result in Fig. 5 demonstrate that the neural network was able to achieve high training accuracy 96.10% with a final validation accuracy of 87.5%. The gap between both

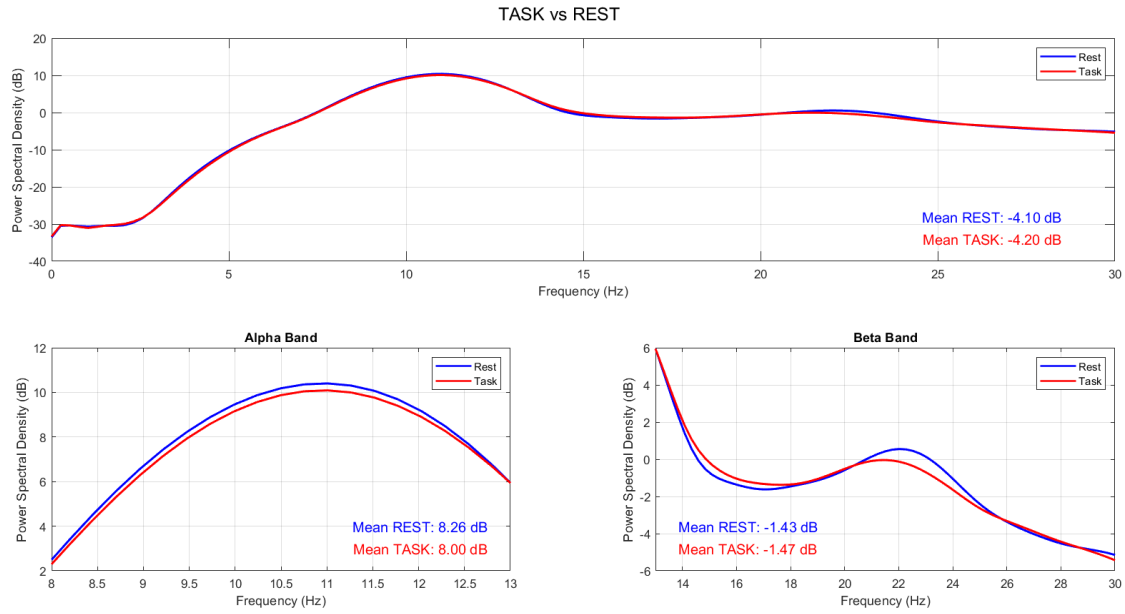


Fig. 4. Periods of hand grasp experiment.

accuracy is relatively low, suggesting there is no overfitting occurred. Both training and validation losses decreased consistently. This also suggest that the model is well-generalized while avoiding overfitting and captures underlying data patters rather than relying solely on memorization of the dataset. Minor variations are observed in the accuracy curve. However, it demonstrates a steady upward trend. This indicates that the model is learning at a consistent pace without experiencing gradient vanishing issues.

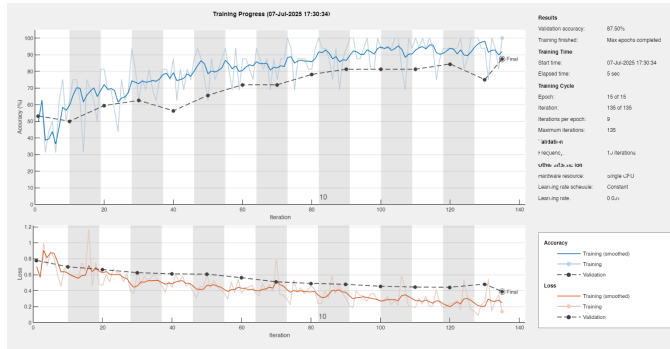


Fig. 5. Neural network training.

The model achieved an accuracy of 81.25% in testing. As illustrated in the Fig. 6, there is a small number of misclassifications still observed. Nonetheless, overall performance is at a reliable level of predictive capability. In the context of BCI systems where signal variability and noise are common, this level of accuracy is considered satisfactory and sufficient to support fundamental interaction tasks, such as binary motor command classification.

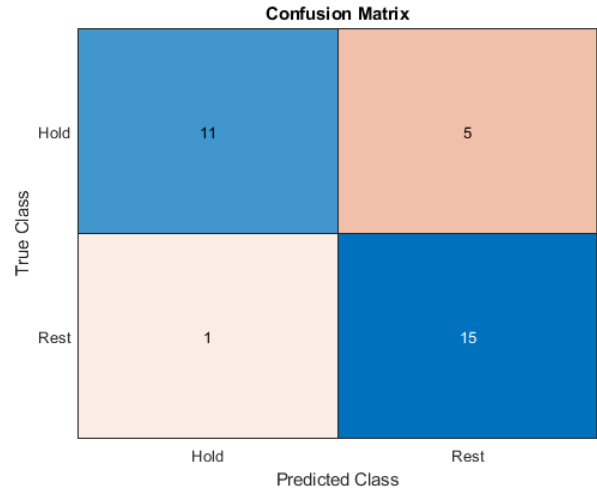


Fig. 6. Confusion matrix of neural network model testing.

Despite the promising performance, the current model still exhibits several limitations. One notable concern is the relatively small dataset used during training and testing which may limit the model's ability to generate across environmental conditions. The presence of a few misclassified samples suggests that the model could be sensitive to subtle signal fluctuations or noise, especially in real-world BCI applications. To enhance robustness, future work should focus on expanding the dataset by including more session repetitions and variations in signal acquisition settings. Additionally, improvements in signal processing methods may further refine the input features and boost classification performance. By combining richer data

with more effective preprocessing strategies, the system could achieve greater reliability and applicability in diverse real-world scenarios.

IV. CONCLUSION

This study demonstrated the feasibility of classifying hand grasping states using power spectral density features extracted from signals denoised with the Discrete Wavelet Transform, as part of the development of a low-cost, open-source brain-computer interface system. EEG signals were recorded using affordable hardware, processed through discrete wavelet transform to isolate the alpha and beta frequency bands, and further analyzed using power spectral density with the Welch method. A neural network classifier was successfully trained on these features, achieving a promising accuracy of 81.25% with stable generalization performance across validation and testing data.

Analysis results showed that the reduction in power spectral density (PSD) between conditions did not reach statistical significance, which may indicate limited engagement in motor imagery or the influence of confounding factors such as movement-related artifacts. This suggests that the current paradigm may not fully capture distinct neural patterns for the targeted states, potentially reducing classification accuracy. Future work could address this by refining task instructions to promote stronger motor imagery, incorporating hybrid paradigms that combine motor imagery with minimal movement, and applying advanced artifact removal techniques. In addition, increasing dataset size, optimizing preprocessing strategies, and exploring additional deep learning models could further enhance robustness and adaptability across users. These findings support the ongoing effort toward accessible and effective BCI technologies, particularly for rehabilitation applications.

REFERENCES

- [1] J. R. Wolpaw and E. W. Wolpaw, "Brain-computer interfaces," Oxford University, Tech. Rep., 2012.
- [2] K. Nakayashiki, M. Saeki, Y. Takata, Y. Hayashi, and T. Kondo, "Modulation of event-related desynchronization during kinematic and kinetic hand movements," Tech. Rep., 2014. [Online]. Available: <http://www.jneuroengrehab.com/content/11/1/90>
- [3] S. K. Satapathy, S. Dehuri, A. K. Jagadev, and S. Mishra, *Chapter 1 - Introduction*. Academic Press, 2019, pp. 1–25. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/B9780128174265000016>
- [4] H. Yuan, C. Perdoni, and B. He, "Relationship between speed and eeg activity during imagined and executed hand movements," *Journal of Neural Engineering*, vol. 7, p. 026001, 4 2010.
- [5] T. Pistohl, T. S. B. Schmidt, T. Ball, A. Schulze-Bonhage, A. Aertsen, and C. Mehring, "Grasp detection from human ecog during natural reach-to-grasp movements," *PLoS ONE*, vol. 8, 1 2013.
- [6] A. K. Engel and P. Fries, "Beta-band oscillations-signalling the status quo?" pp. 156–165, 2010.
- [7] H. Tan, A. Poghosyan, A. Anzak, K. Ashkan, M. Bogdanovic, A. L. Green, T. Aziz, T. Foltynie, P. Limousin, L. Zrinzo, and P. Brown, "Complementary roles of different oscillatory activities in the subthalamic nucleus in coding motor effort in parkinsonism," *Experimental Neurology*, vol. 248, pp. 187–195, 10 2013.
- [8] A. Muhawarman, "Cegah stroke dengan aktivitas fisik," 10 2024. [Online]. Available: <https://kemkes.go.id/id/trilis-kesehatan/cegah-stroke-dengan-aktivitas-fisik>
- [9] A. A. R. Putri, M. Siddik, D. D. Sanyoto, A. Hayati, P. S. K. P. Sarjana, and F. K. dan Ilmu Kesehatan, "Perbedaan gangguan fungsi tangan ditinjau berdasarkan karakteristik pasien pasca stroke di poli rehabilitasi medik rsud ulin banjarmasin," Tech. Rep.
- [10] A. A. Portnova, G. Mukherjee, K. M. Peters, A. Yamane, and K. M. Steele, "Design of a 3d-printed, open-source wrist-driven orthosis for individuals with spinal cord injury," *PLoS ONE*, vol. 13, 2 2018.
- [11] S. M. Hermawan and H. Wihardja, "Hubungan karakteristik individu dan riwayat stroke dengan kekuatan otot pada pasien stroke," *Dunia Keperawatan: Jurnal Keperawatan dan Kesehatan*, vol. 8, 2020.
- [12] P. Sovka, A. Stancak, J. Štastný, and A. Stančák, "Eeg signal classification: Introduction to the problem," Tech. Rep., 2003. [Online]. Available: <https://www.researchgate.net/publication/26511576>
- [13] N. Bajaj, "Wavelets for eeg analysis," Tech. Rep. [Online]. Available: www.intechopen.com
- [14] P. Manhas and S. Thakral, "Image processing by using different types of discrete wavelet transform," Tech. Rep., 2018. [Online]. Available: <https://www.researchgate.net/publication/326572441>
- [15] S. Sharma, S. Sharma, and A. Athaiya, "Activation functions in neural networks," *International Journal of Engineering Applied Sciences and Technology*, vol. 4, pp. 310–316, 2020. [Online]. Available: <http://www.ijeast.com>